How the exam runs, and how it is scored

Twenty-five minutes: twenty of examination, then five in which the examiner completes the form. Everyone gets the same four segments in the same order. Before the clock starts: ID check, the recording begins, and your four notebooks and your code are open on your own screen.

THE TWENTY MINUTES

1	**Showcase**	~3 min	you	Pick one result from your four assignments that you are proud of. Show it and explain it. You may prepare this. The follow-up question is not preparable.
2	**Deep dive**	~8 min	examiner	One assignment, drawn in advance and revealed only now, so prepare all four. Anchored in your own code and results through your code map.
3	**Verification and validation**	~6 min	examiner	The tests behind the work, sampling the assignments that were not your deep dive. What each test proves, and what it cannot.
4	**Extension**	~3 min	examiner	Adaptive. A strong exam is pushed higher; a stumble is given a second sounding on a fresh question. Everyone gets this segment.

Once per exam the examiner may ask for a live action: run a test, re-plot a result, or predict the effect of a parameter change before reasoning it through.

DEPTH LEVELS

the examiner escalates until the answers give out, and your grade comes from the ceiling you reach, not from time survived

L1	**describe**	what you did, what the plot shows, what the function computes
L2	**explain**	why it is correct, why that algorithm, what the test verifies
L3	**predict and connect**	what changes if, why the physics comes out this way, quantitative about your own numbers
L4	**critique and design**	limits of model and method, design a new test, reach beyond the handed material

THE FOUR DIMENSIONS

each scored 0 to 4 against these anchors

	0	1	2	3	4
D1 Implementation command *your own code*	cannot locate or explain your own code, even when prompted	locate it, but describe it only procedurally, and err on how it works	explain the mechanics correctly, with prompting	explain fluently unprompted, and reason about hypothetical changes	discuss design alternatives, costs and limits, beyond what the task required
D2 Verification and validation *the tests behind it*	cannot say how the work was verified	name the tests, but not what they demonstrate	explain what each test verifies and what a pass means, with prompting	explain what a test can and cannot catch, predict outcomes under change, justify tolerances	design new tests on the spot, critique against the literature, separate verification from validation
D3 Physical interpretation *your own results*	cannot state what your results show	read plots literally, with no connection to the physics	connect results to the underlying physics, with prompting	reason quantitatively about your own numbers and predict parameter effects	relate results to literature and experiment, discuss model limits and regime of validity
D4 Methods landscape *and transfer*	no grasp of what the methods are	describe MD, DPD and DEM only as the assignments	state the essential differences, with prompting	choose a method for a new problem with correct justification, and know what transfers and what breaks	discuss coarse-graining rationale and method limits, and connect to the wider simulation landscape

FROM THE FOUR SCORES TO A GRADE

S is the sum over the four dimensions, 0 to 16

S	6 – 7	8 – 9	10 – 12	13 – 14	15 – 16
Grade	**6**	**7**	**8**	**9**	**10**

A fail if any dimension scores 0, or if D1 scores 1 or less, or if S is 5 or less. D1 is the backbone: a student who cannot explain their own submitted work does not pass on the strength of the other three.

The examiner may depart from this mapping by at most one grade point, with a written justification on the form. Your final mark is the oral grade multiplied by the weights of the reports you handed in, 0.25 each, so with all four in, your final mark is your oral grade. The grade is registered within one working day.